

Lab 04 - Equivalent Networks and Superposition

Introduction

For this lab you will be exploring some of the circuit solving techniques you have been studying in lecture. Specifically, you will simulate and build three DC circuits to experimentally test these concepts. The first procedure will verify Thévenin's equivalent circuit. The second procedure will verify the superposition principle. After you verify that the superposition principle works, you will also test the conditions under which superposition can be applied. In this lab, you will also learn more about how LTSpice can be used to model a circuit.

Equivalent Circuits

As you have learned in lecture, circuits can be transformed to equivalent circuits to ease in analysis. In this lab, we will focus on the Thévenin equivalent circuit shown below in Figure 4.1a. Remember that you can always use a source transformation to move from the Thévenin equivalent to the Norton Equivalent circuit shown in Figure 4.1b and vice versa.

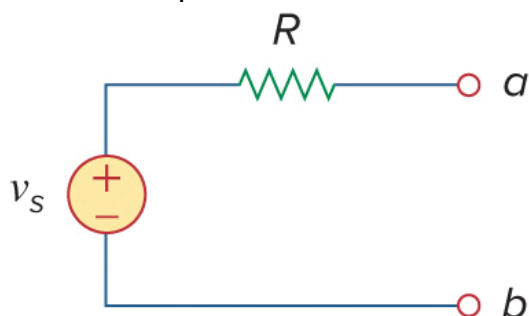


Figure 4.1a: Thévenin Equivalent Circuit

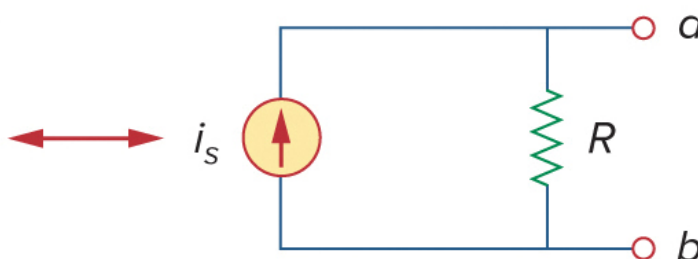


Figure 4.1b: Norton Equivalent Circuit

Superposition

The Superposition principle was covered in lecture. However, if you need more detail on it before beginning this lab, please read the appropriate sections of your text. Simply put, superposition holds that if you have a linear circuit with multiple independent sources, you can solve the circuit by solving a circuit with each source in the circuit alone. Then, summing the individual results (with respect to polarity) should give you the same result as if you solved the circuit with all independent sources present. In the experimental portion of this lab, you will be testing this principle.

Linear versus Nonlinear Devices

So far, you have encountered mainly resistors when building circuits. Resistors are generally modeled as linear devices in this course. However, there are other devices that are non-linear. For example, operational amplifiers (Op-Amps) have both a linear and a non-linear range of operation. In this lab, while testing the superposition principle for a non-linear circuit, you will have a diode in the circuit (see figure 4.2).

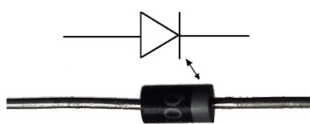


Figure 4.2: Diode and Its Representation

You will not study diodes in detail in this course, but you will extensively study and use diodes in other courses. However, there will be occasions in the lab this semester where you will use diodes in your circuits. Therefore, it is a good idea to become familiar with them. If you have not studied diodes before, the important information you need to know is that current only flows through a diode from the positive (anode) terminal to the negative (cathode) terminal. Current cannot flow from the negative side of a diode to the positive side of a diode. Therefore, you can also consider a diode like a switch. When current is pushed from the positive to the negative terminals of a diode, the switch is closed. When current tries to flow from the negative to the positive side of a diode, the switch opens and no current is allowed to flow. Also, when current is flowing through a diode, there is a voltage drop of approximately 0.7V across the diode (measured from the anode to the cathode).

Viewing and using the Netlist file from LTSpice

After you create a schematic in LTSpice, you have the option of viewing the Netlist. If there is a problem with your circuit you do not understand, checking the LTSpice netlist (or the circuit file) can help you identify problems. If you are using LTSpice, go to the View menu and click on “SPICE Netlist” to view the netlist. A new file (XXX.net) will come up in an editor for you to view. Every time you change your schematic, a new netlist is created.

For example, the schematic in Figure 4.3 was drawn in LTSpice, and the simulation was run to find the voltages and currents on the schematic.

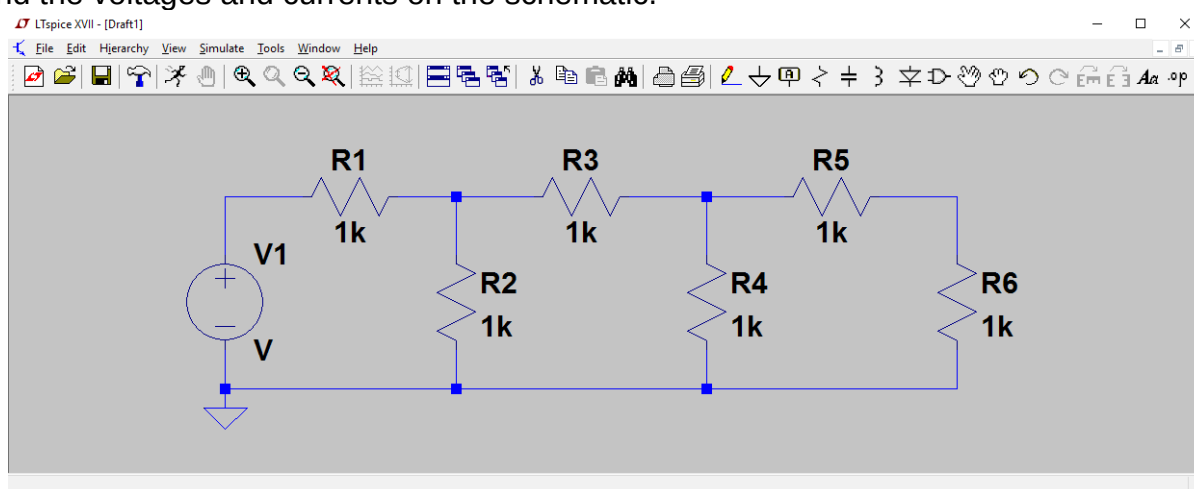


Figure 4.3 – LTSpice Schematic for a Sample Circuit

The netlist that is created from this schematic is:

```
V1 N001 0 V
R1 N002 N001 1k
R2 N002 0 1k
R3 N003 N002 1k
R4 N003 0 1k
R5 N004 N003 1k
R6 N004 0 1k
```

Note how each part is given by its type and name (R_n), the two nodes it is connected to (N_{xxx} and N_{yyy}), and the value of the component. For example, the resistor labeled $R3$ is connected to node 3 ($N0003$) and node 2 ($N0002$) and is a $1k\Omega$ ($1k$) resistor. The ground node is indicated by 0. LTSpice will not necessarily label the nodes the same way you would. However, from the list above, you should be able to draw the circuit.

Working with schematics in LTSpice can be very helpful. However, working with the netlist and circuit files that LTSpice uses to generate output can help you understand more about how circuit simulation works.

Part 1: Calculations and PreLab Simulations

1. Refer to the circuit in Figure 4.4. Simulate the circuit in Figure 4.4 in LTSpice. **Print a screenshot of your circuit schematic with all voltages and currents shown (DC operating point simulation).**

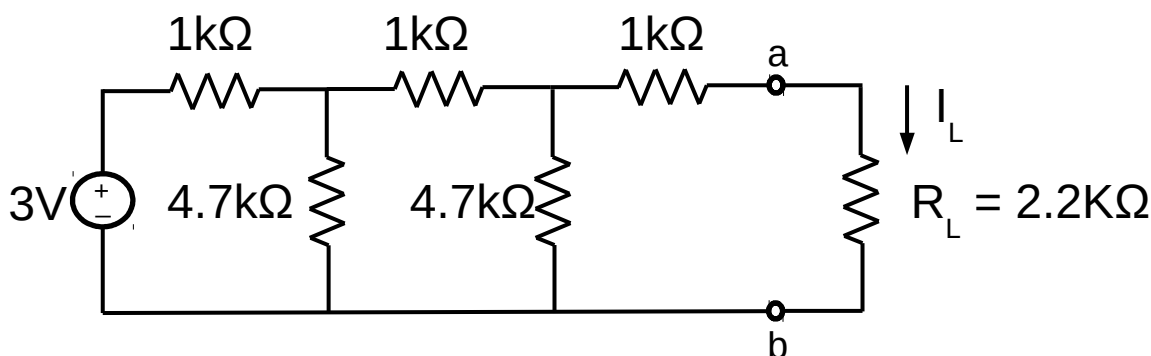


Figure 4.4

2. Calculate V_L (voltage across R_L) for the circuit in Figure 4.5 using superposition. **Fill out the column entitled "Calculation" in Table 4.1.**
3. In addition, simulate the circuit in LTSpice. **Print the schematic with all voltages and currents displayed.** You can test the superposition principle by running 2 more simulations: one with V_1 replaced with a short and one with V_2 replaced with a short. You do not need to print all of the schematics to turn in. However, you should run all the simulations and **fill out the column entitled "Simulation" in Table 4.1.**

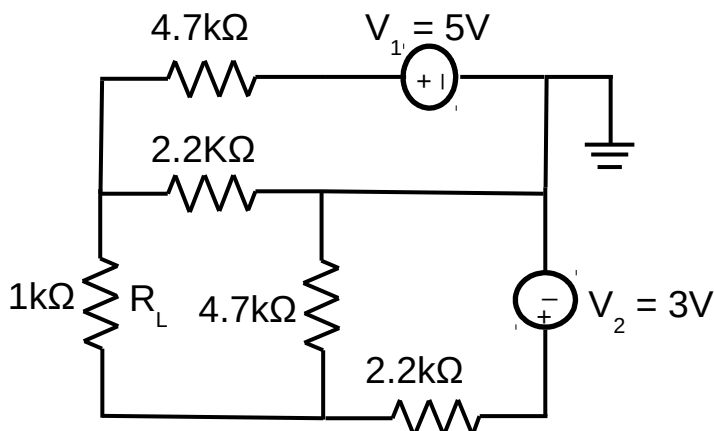


Figure 4.5

4. Simulate the circuit in Figure 4.6. You will want to use part 1N4148 for the diode in LTSpice. **Print the schematic with all voltages and currents displayed.** Again, test the superposition principle by running 2 more simulations as above. Again, you do not need to print all of the schematics to turn in, but run the simulations and **fill out the column entitled “Simulation w/Diode” in Table 4.1**

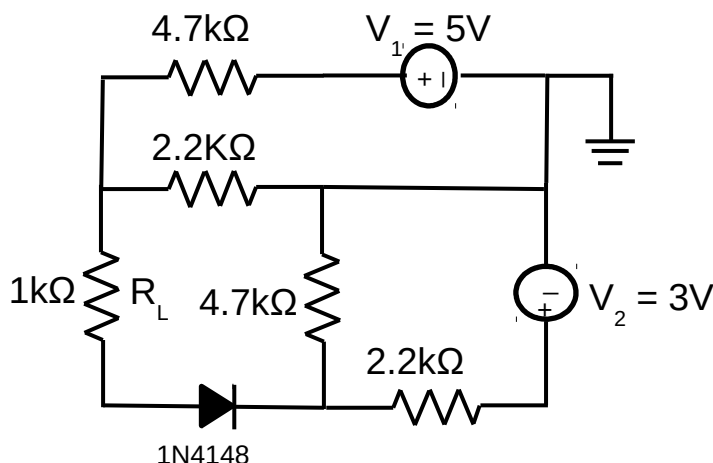


Figure 4.6

V_L (across $R_L/1k\Omega$)	Calculation	Simulation	Simulation w/Diode
V_1 & V_2 Present (A)			
V_1 only			
V_2 only			
Add line 2 & 3 (B)			
% Δ between A and B			

Table 4.1: Calculations & simulations with and without a diode

For which simulation does the superposition principal hold?

Remember to bring your schematics and a filled out Table 4.1 from Part 1 to the in-lab so you can use them to do the experiments.

Part 2: In-Lab Experiments

Parts needed:

- Electronics Trainer
- Three 1.0 k Ω resistors
- Two 2.2 k Ω resistors
- Two 4.7 k Ω resistors
- One 1N4148 diode
- A selection of colored 22 gauge connection wires

It is a good idea to verify all your component values before you begin doing an experiment.

Task 1 - Verify Thévenin's Equivalent

Construct the circuit of Figure 4.4. Measure the exact voltage provided by your source before you connect it to the rest of the circuit and do not connect the voltage supply to the circuit until you have completed all the other connections. **Double check your connections!** Make the following measurements on this circuit:

1. Use your voltmeter to measure the voltage (V_L) on the load resistor (R_L) for two different values of R_L . Choose any two different values, but not 2.2k Ω .
2. Now use your voltmeter to measure the V_L when the $R_L = 2.2\text{k}\Omega$.

Using the measurements you made in item 1 above, calculate the Thévenin equivalent of the circuit. **In your lab report, be sure to explain how you came up with your Thévenin equivalent model from the two voltage measurements taken in step 1.** Based on your measured Thévenin equivalent circuit, what should be the voltage across a 2.2k Ω load resistor? How does this compare with the actual voltage you measured? Discuss the reasons for any differences you see. Record your measurements and calculations.

Task 2 – Verify the Superposition Principle

Construct the circuit of Figure 4.5. You can use your batteries for the 3V source and the variable voltage source as the 4V source. Make the following measurements with your voltmeter:

1. Measure the voltage across the 1k Ω resistor and record in your lab notebook
2. Remove V_2 (3V) from the circuit and replace it with a short circuit. Measure the voltage across the 1k Ω resistor. Call it $V_{L,1}$ and record it in your lab notebook.
3. Put V_2 back in the circuit and remove V_1 (4V) and replace it with a short circuit. Measure the voltage across the 1k Ω resistor. Call it $V_{L,2}$ and record it in your lab notebook.

Does the principle of superposition apply to this circuit? Compare these measured values with the theoretical values in Table 4.1 and discuss any significant differences that you see.

Task 3 – Check the Superposition Principle validity for a Non-linear Device

For Task 3, repeat the procedure from Task 2, but use the circuit from Figure 4.6, which added the diode. Be sure to place the diode in the circuit with the polarity indicated in Figure 4.6 so that the current will flow in the correct direction.

1. Measure and record the voltage across the 1k Ω resistor with both V_1 and V_2 present.
2. Measure and record the voltage across the 1k Ω resistor with only V_1 present
3. Measure and record the voltage across the 1k Ω resistor with only V_2 present

Table 4.2: Task 2

Parameter	Calculated	Simulated	Measured	%diff (C v M)	% diff (S v M)
V_L					
$V_{L,1}$					
$V_{L,2}$					
$V_{L,1} + V_{L,2}$					

Table 4.3: Task 3

Parameter	Simulated	Measured	% diff (S v M)
V_L			
$V_{L,1}$			
$V_{L,2}$			
$V_{L,1} + V_{L,2}$			

Lab Report

Submit a lab report based on the lab report template provided on Blackboard.